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Ablation of a maternal *Cryptosporidium* mRNA-binding protein results in sterile sporozoites

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ABSTRACT Infection with *Cryptosporidium* is a leading cause of diarrheal disease and early childhood mortality. This apicomplexan parasite undergoes asexual and sexual replication within the same host, and recent studies have shown an intrinsic developmental program of obligate transition to male and female gametes and sex. While factors were identified that control male fate and development, how female gene expression is orchestrated remains largely unknown. Here, we use the *Cryptosporidium* Single Cell Atlas to discover an RNA-binding protein (F-RBP) as one of the earliest markers of female identity in *C. parvum*. Reporter parasites engineered based on this gene allowed us to calibrate transcriptional pseudotime against the real time of female development, revealing a significant window of transcriptional fate ambiguity. While F-RBP is an early transcript, the protein persists throughout female development and into the zygote. Conditional ablation of the F-RBP gene showed it to be dispensable for sex determination and early female development *in vitro*. However, the gene is essential *in vivo,* and its loss results in rapid cure. Cell biological experiments link this loss to the production of oocysts that release sporozoites incapable of host-cell invasion. F-RBP binds transcripts highly expressed in the female gamete, enriched for a sequence motif recognized by a Y-box protein family member, and forms mRNA-protein complexes in late females akin to processing or *P* bodies. We propose F-RBP's essential role to be in the regulation of long-term homeostasis of maternally inherited RNA required for sporozoite infectivity.

IMPORTANCE The parasite *Cryptosporidium* is responsible for millions of cases of severe diarrheal disease in young children, and the infection is both a consequence of and a risk factor for chronic malnutrition. There are no vaccines available, and a single approved treatment is largely ineffective. Infection relies on ingestion of the oocyst, an environmentally stable spore, which contains an invasive form of the parasite, the sporozoite. Oocysts and sporozoites are the products of parasite sex and the fusion of female and male *Cryptosporidium* parasites. Here, we report the critical role of an RNA-binding protein in female mRNA homeostasis. Maternal transcripts orchestrate the complex process of sexual replication and oocyst formation. Loss of the protein results in spores that fail to initiate the next round of the life cycle, and conditional ablation using a small-molecule trigger quickly cures infected animals, demonstrating the essential role of parasite sex in continued infection.

KEYWORDS *Cryptosporidium*, apicomplexan parasites, RNA-binding proteins, intracellular parasites, diarrhea, female gamete, parasite sex

Sexual reproduction is a hallmark of eukaryotic life. Two distinct and often differently sized cells contribute to this process: the male cell (sperm) and the female cell (egg). The sperm's primary role is to deliver its genetic material. The egg is responsible for enabling male recognition to allow fertilization, prevention of polyspermy, supporting early embryogenesis, and often storing and providing the metabolic reserves to fuel

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these activities. Egg cells harbor mRNAs that encode proteins needed throughout egg development and early embryogenesis (1, 2). Carrying these mRNAs allows the fertilized egg to execute a series of rapid transitions without the need for transcription. Multicellular and unicellular eukaryotes alike use RNA-binding proteins to ensure that these messages are stored, protected, and transcribed exactly when needed in development (1–4).

Apicomplexan parasites similarly rely on sexual reproduction. *Cryptosporidium* goes through a defined series of asexual replication cycles that culminate in the formation of the sperm and egg equivalents, the male and female gametes, respectively (5). The male gamete fertilizes the female gamete, giving rise to the oocyst, a spore-like stage that can transmit the infection to a new host or remain within the original host to continue the infection.

Akin to other female reproductive cells, the *Cryptosporidium* female gamete prepares for different events at distinct developmental stages. First, the female gamete commits to and initiates the female genetic program; then, the parasite grows significantly and stockpiles metabolites, proteins, and mRNA. *Cryptosporidium* is a haplont and will enter meiosis immediately after pronuclear fusion, and the female gamete carries the required machinery (6–10). Maternal mRNA is also required for the assembly of the crystalloid body (7), the synthesis of storage polymers (11), sporozoite assembly (12), and the biogenesis of the oocyst wall (13). The oocyst wall is composed of glycans, specialized lipids, and proteins, including the *Cryptosporidium* oocyst wall proteins (COWPs) (6, 13–16). Given suitable conditions, the oocyst can remain dormant for months (17) and, upon ingestion, release motile sporozoites that invade the intestinal epithelium to colonize a new host.

The *Cryptosporidium* life cycle has an obligate switch toward sex (5, 8, 18), and sex is required to sustain continued infection (6, 7, 9–11, 18, 19). In standard cell culture, gametes are formed, but fertilization does not occur, and growth ceases after 2 days (11). Organoid culture allows for oocyst production, though not at levels observed in mice (19). Remarkably, preventing parasite sex will rapidly halt the infection in the same host (6).

Apetala 2 (AP2) transcription factors mediate developmental changes in a variety of apicomplexan parasites and are important regulators of gene expression, especially during sexual development (20–22). The female gamete expresses three of these transcription factors (6). One has been implicated in the development of the sporozoite-specific organelle, the crystalloid body (7), while another has been linked to female gamete development *in vitro* (23). However, all three of these transcription factors are transcribed at a similar late point in development, providing only limited time resolution (6).

Here, we characterize a female-specific RNA-binding protein, F-RBP. F-RBP preferentially binds mRNAs that are upregulated in female development, and the absence of F-RBP results in female sterility and sporozoites incapable of host-cell infection.

RESULTS

Female identity is transcriptionally evident only at the halfway point of gamete development

We mined the *Cryptosporidium* female-exclusive single-cell transcriptome (6) for potential regulators of female gene expression, focusing on genes expressed very early and exclusively during female development (12). We identified two genes: cgd4_130, encoding a putative RNA recognition motif (RRM) and K homology (KH) RNA-binding protein, and cgd4_1360, a zinc finger domain-containing protein (referred to as F-RBP and F-ZF, respectively; Fig. 1A). Single-cell RNA profiling has provided significant insight into the relative sequence of gene expression in *Cryptosporidium,* but we do not understand how the pseudotime derived from these analyses translates into the actual time of parasite development. To synchronize these two clocks and to monitor development over time, we constructed a fluorescent reporter strain that combined the

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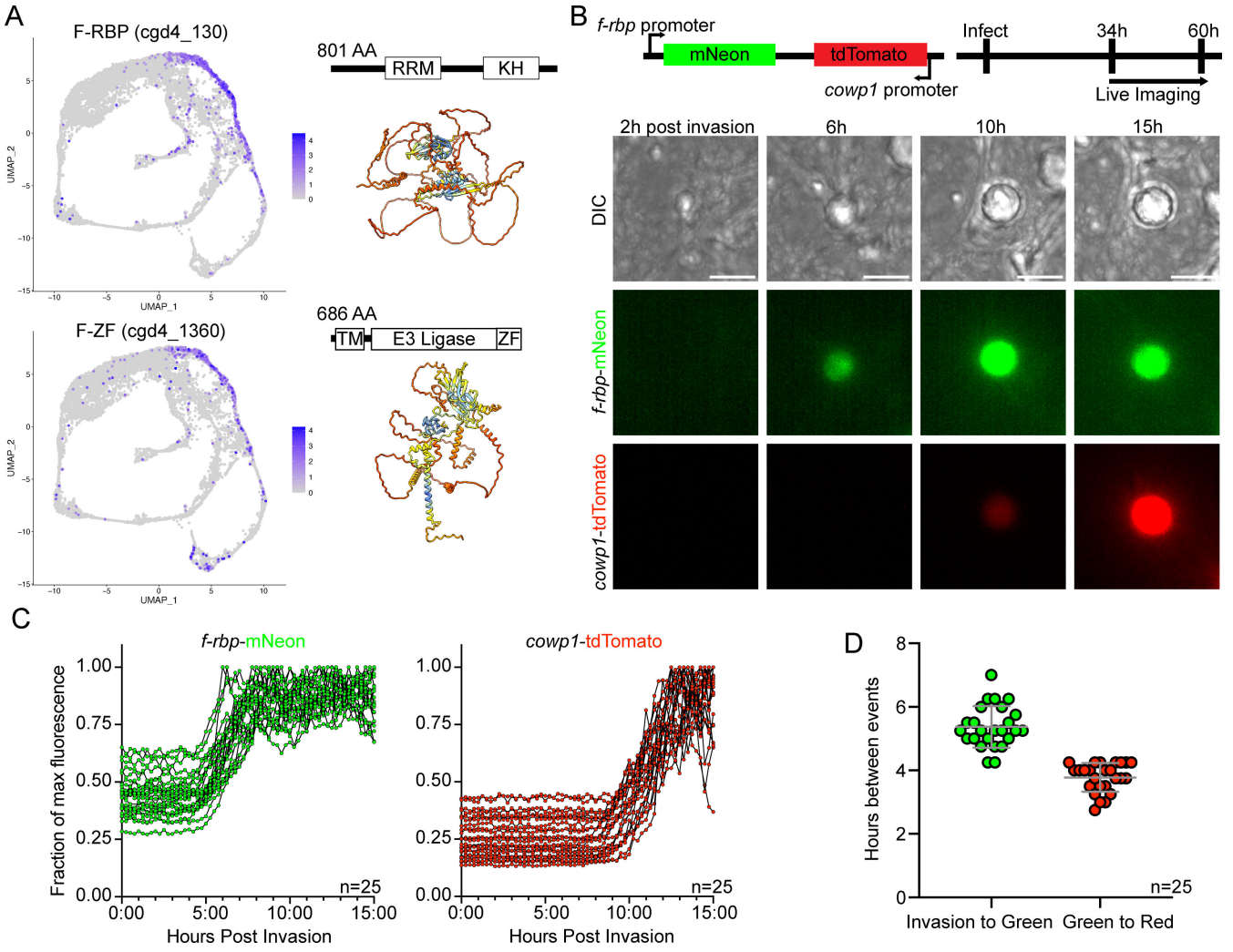


FIG 1 F-RBP expression defines early female gamete development. (A) Transcript expression mapped onto the Single Cell Atlas (6), domain organization, and AlphaFold structure predictions of F-RBP (top) and F-ZF (bottom). (B) Construct and experimental design for the female reporter live-imaging assay (top). Representative fluorescence and differential interference contrast (DIC) images of a female gamete over time. Scale bar, 5 µm. (C) Fluorescence intensity plotted as a fraction of maximum fluorescence over time for *f-rbp*-mNeon and *cowp1*-tdTomato. Each continuous line represents a single tracked parasite. n = 25. (D) Time measured between the indicated events for each of the 25 gametes. "Invasion to green" represents the time from invasion to the first detection of mNeon. "Green to red" indicates the time between the first detection of mNeon and the first detection of tdTomato. First detection was defined as the time when fluorescence intensity exceeded the mean background by five times the SD. The mean and SD are indicated in gray. N = 25.

f-rbp promoter to control expression of the green fluorescent protein mNeon and the promoter of the previously characterized *cowp1* (11) to control expression of the red fluorescent protein tdTomato (Fig. 1B). Stable transgenic parasites were selected with paromomycin, and genome modification was validated by PCR mapping of genomic DNA isolated from oocysts purified from the feces of infected mice (Fig. S1A) (24).

Transgenic parasites were used to infect the human adenocarcinoma cell line HCT-8 seeded into glass-bottom chamber slides and subjected to time-lapse imaging after 34 h of growth (see Materials and Methods for technical detail). At this point, parasites egress from their final round of asexual replication, followed by reinvasion and differentiation into gametes (6, 18). We detected no fluorescence in either channel for invading merozoites (Fig. 1B and C; Movie S1). When tracking the resulting intracellular stages, mNeon driven by the *f-rbp* promoter was detectable first, followed later by *cowp1*-driven tdTomato. At the end of the observation period, we expected to see these parasites expressing both markers, which we observed (Fig. 1B; Movie S1). We recorded the

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development of 25 female gametes. Parasites were segmented in each frame using the DIC images as a mask, and we measured and plotted the fluorescence intensity for both reporters over time. We found that the timing of reporter expression was remarkably consistent (Fig. 1C). After the parasites invade, it takes 5 h and 20 min (SD, ±40 min) for the *f-rbp* promoter to turn on, marking them as female. It then takes another 3 h and 45 min (±30 min) to detect *cowp*1-driven fluorescence (Fig. 1D). We thus note a substantial phase in which the life cycle stage remains measurably transcriptionally undifferentiated. After this period, gametogenesis proceeds through precisely timed waves of female gene expression, with *cowp1* expression occurring at 9 h during the roughly 12-h period of gamete development.

F-RBP and F-ZF proteins are translated early in female development

To assess the levels of F-RBP and F-ZF proteins over the course of parasite development, we introduced a DNA cassette encoding a 3×HA epitope tag to modify the C terminus of each protein at the native gene locus (Fig. S1B and C). HCT-8 cultures were infected with the resulting transgenic parasites and processed for immunofluorescence microscopy at different time points. No labeling was observed for F-RBP-HA or F-ZF-HA after 24 h of infection, in accordance with our transcriptomic and reporter results (Fig. 2A through C). F-RBP-HA was readily observed in cultures starting at 42 h post infection, accounting for approximately 30% of all parasites detected by counterstaining with *Vicia villosa* lectin (VVL, Fig. 2A and C). This proportion increased to 50%–60% of parasites at 48 h post infection and remained so at 72 h, consistent with the proportion of female parasites previously reported for these timepoints (11). F-ZF-HA was detected in a smaller fraction of parasites at 42 h (0%–15%), increasing to ~30% of the population at 48 h and 50% at 72 h post infection (Fig. 2B and C). For both lines, HA+ parasites were observed to have a single nucleus, typical for the female gamete (11, 18), and both proteins were consistently detected in parasites positive for the female marker DMC1 (7, 8). DMC1 is expressed later in female development, and we therefore detect F-RBP-HA+ parasites that are not yet expressing DMC1 (Fig. S2A). All DMC1 + parasites also stained F-RBP-HA+ (Fig. S2A).

F-RBP is present in zygotes but is absent in oocysts

While female gametes form in HCT-8 cultures, they are not fertilized, precluding the observation of subsequent development (11). To investigate F-RBP expression *in vivo,* we infected C57BL/6 *ifng* $^{-/-}$ mice and, at peak infection, sacrificed the animals, resected the ileum, and performed immunofluorescence on histological sections. We noted the presence of F-RBP in female gametes, identified by their single, large nucleus (Fig. 2D). We also detected F-RBP in zygotes, which carry both a large female nucleus and a smaller male nucleus, demonstrating that this protein persists from early gametogenesis through fertilization and the maternal-to-zygote transition (Fig. 2E). We next sought to determine the fate of F-RBP in oocysts. This is technically complicated by the oocyst wall, which is not readily breached by detergent permeabilization, leading to inconsistent antibody access and staining. To overcome this, we engineered a parasite strain in which F-RBP was translationally fused to eGFP, thus allowing us to use endogenous fluorescence to monitor the presence of F-RBP protein (Fig. S1D).

While these parasites showed bright fluorescence in female stages at 48 h *in vitro* (Fig. 2F), no fluorescence was detected in oocysts by either microscopy or flow cytometry (Fig. 2G). We conclude that F-RBP is expressed early in female development, maintained through gametogenesis and fertilization, but degraded by the time the oocyst has matured.

Genetic ablation of F-RBP and F-ZF

To assess the importance of F-RBP or F-ZF for female development and parasite life cycle completion, we attempted to ablate each gene by homologous recombination. We readily achieved this for F-ZF, and oocysts were abundantly shed from mice infected

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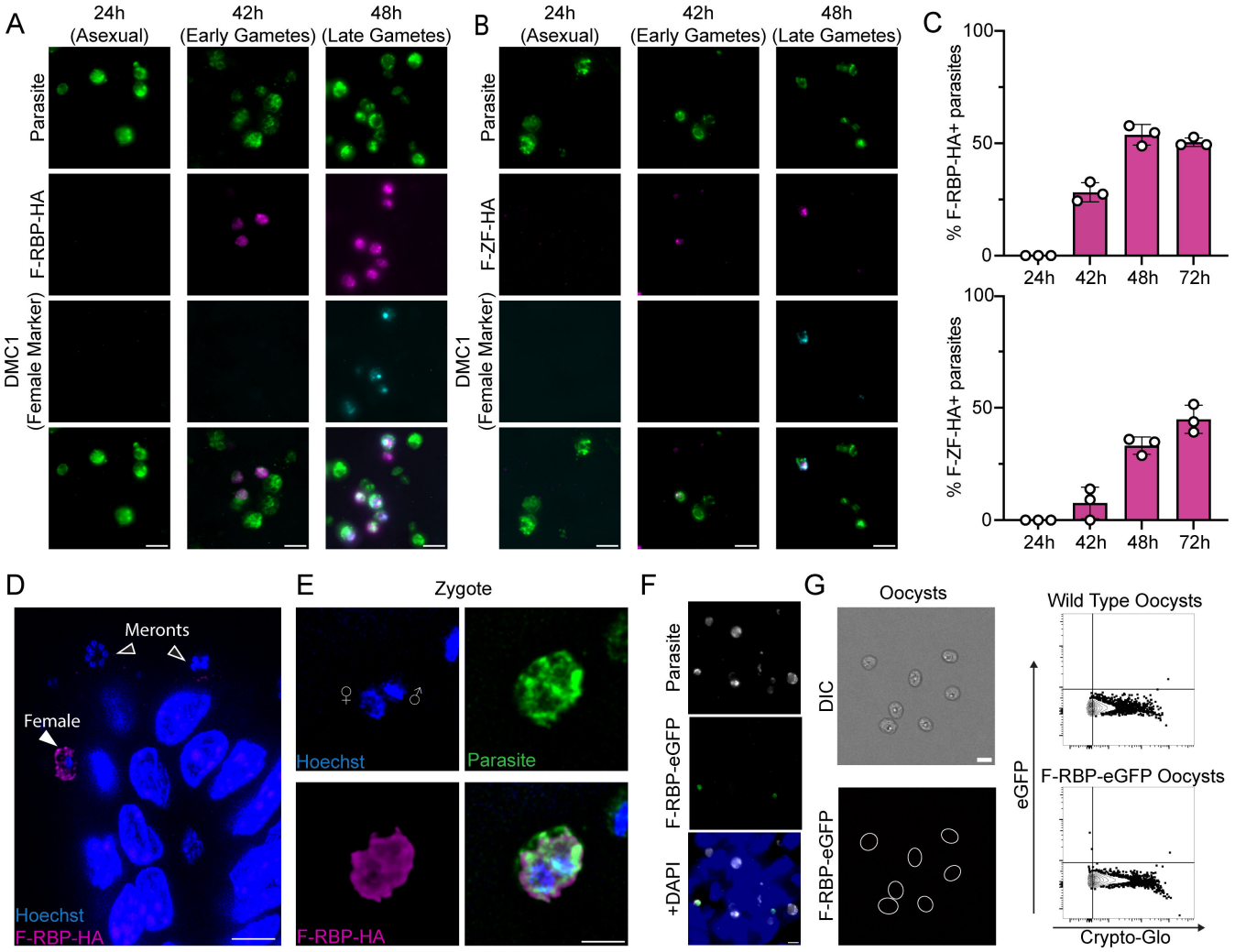


FIG 2 F-RBP protein is detectable from the early gamete to the zygote but is absent from the oocyst. Immunofluorescence time course of (A) F-RBP-HA and (B) F-ZF-HA in HCT-8 tissue culture. Scale bar, 5 µm. Representative images from three independent experiments. (C) Quantification of HA-positive parasites for F-RBP-HA (top) and F-ZF (bottom). A total of 3,438 parasites were counted for F-RBP quantification and 3,920 parasites were counted for F-ZF quantification. For both protein time courses, three independent experiments, with three coverslips per experiment, were counted. Data from one representative experiment are shown. (D) F-RBP-HA staining in a tissue section from an infected mouse intestine. Different parasite stages are highlighted. Scale bar, 5 µm. (E) Detail from a tissue section showing a zygote (recognized by the presence of both a female [large] and a male [small, bullet-shaped] nucleus). Note that the zygote is stained for F-RBP-HA. Scale bar, 2 µm. (F) F-RBP-eGFP parasites in HCT-8 cells. Scale bar, 5 µm. Note the female parasites showing eGFP fluorescence. (G) F-RBP-eGFP oocysts harvested from infected mice and examined by microscopy (left) or flow cytometry (right). Flow cytometry results are representative of two independent experiments. F-RBP-eGFP oocysts show no fluorescence above the wild-type control. Scale bar, 5 µm.

with validated ΔF-ZF parasites, indicating that this gene is dispensable for life cycle completion (Fig. S3A and B). In contrast, parasites were not recovered in three separate attempts to ablate the C-terminal portion of F-RBP, which included the KH domain (Fig. S3C). We found F-ZF to be dispensable, and loss of the gene did not result in obvious phenotypic differences, while F-RBP was refractory to our knockout attempts. We thus focused the subsequent experiments on F-RBP and its likely essential function.

To test rigorously whether F-RBP is essential, we sought to establish a conditional knockout using rapamycin-inducible Cre recombinase (6, 7, 25). However, multiple attempts to insert loxP sites were unsuccessful (Fig. S1E). Figure S1E details the various targeting constructs we explored, leading us to consider the position of the double-strand break. Multiple studies have shown that introducing double-strand breaks in close proximity to the desired site of homologous recombination can enhance efficiency (26).

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We tested this using two guide RNAs to cut immediately upstream and downstream of the region to be replaced by homologous recombination. Alternatively, we used a single guide and designed a repair template for insertion instead of replacement, generating a partially pseudo-diploid floxed locus (see Fig. 3A; and Fig. S1E and F for details on both strategies). The second approach proved successful (Fig. 3B; Fig. S1F). We engineered F-RBP-flox parasites in two *C. parvum* backgrounds, Iowa IIa Bunchgrass and the KVI IId strain (27), and then crossed them with Iowa IIa Bunchgrass parasites carrying a rapamycin-inducible Cre recombinase, yielding F-RBP cKO parasites (25). The IIa background mutant grew poorly, and we therefore focused our efforts on the mutant in the IId background. HCT-8 cultures were infected with this mutant and incubated in the presence of rapamycin or an ethanol vehicle. Rapamycin treatment resulted in complete Cre-mediated excision of the floxed sequence from the chromosomal locus over 48 h, and, as previously described (6, 7), we also noted partial excision in the absence of rapamycin due to leaky Cre activity (Fig. 3C).

F-RBP is dispensable for female gametogenesis and growth *in vitro*

We first measured the impact of F-RBP ablation on parasite growth *in vitro*. HCT-8 cells were infected with either F-RBP-flox parasites (which lack Cre recombinase) or F-RBP cKO parasites and subjected to vehicle or rapamycin treatment. Measuring parasite burden after 48 h using a luciferase assay showed that rapamycin treatment did not inhibit the growth of either strain (Fig. 3D). F-RBP is one of the earliest female transcripts, and we therefore wondered whether it might be required for female development. HCT-8 cell cultures infected with mutant parasites were treated with vehicle or rapamycin, and we measured transcript abundance for DMC1, a meiotic protein exclusively expressed in females (7, 8). There was no significant difference between the two conditions (Fig. 3E). We also assessed female development by IFA using a DMC1 antibody (7, 8) and again found no difference upon loss of F-RBP, as we readily identified parasites expressing DMC1 but lacking F-RBP-HA after rapamycin treatment (Fig. 3F).

F-RBP ablation cures infected mice

We next tested whether F-RBP may be required *in vivo*. Mice were pre-treated with either vehicle or rapamycin for 2 days and infected with F-RBP-flox or F-RBP cKO parasites, and treatment was continued for 10 days. Oocyst shedding was evaluated by measuring luciferase activity in the feces (24). Mice infected with F-RBP-flox parasites shed oocysts robustly and developed severe infection irrespective of treatment (Fig. 3G). The rapamycin-treated group shed more oocysts due to the immunosuppressive activity of the drug, as reported previously (7). While mice infected with F-RBP cKO parasites shed oocysts throughout the observation period when treated with vehicle, rapamycin treatment severely reduced oocyst shedding (>99.7% reduction compared with the vehicle control in both independent experiments) (Fig. 3G). Nanoluciferase assays conducted using intestinal biopsy material showed a similarly reduced parasite burden in the tissue after rapamycin treatment (Fig. 4C). We confirmed this reduction in parasite burden by performing immunofluorescence on sections of intestinal tissue (Fig. 4C). We also tested whether patently infected animals could be cured by F-RBP ablation. Mice infected with F-RBP cKO parasites were split into two groups at the peak of infection. While vehicle-treated animals remained infected, rapamycin-treated animals displayed a rapid decrease in fecal luciferase signal (Fig. S4A). We conclude that F-RBP is dispensable for female development *in vitro* but essential *in vivo* and required for sustained infection.

Oocysts produced by F-RBP-deficient females do not reseed intestinal infection

We next sought to understand at what point loss of F-RBP halts the infection. Mice infected with F-RBP cKO parasites for 6 days were treated with rapamycin or vehicle

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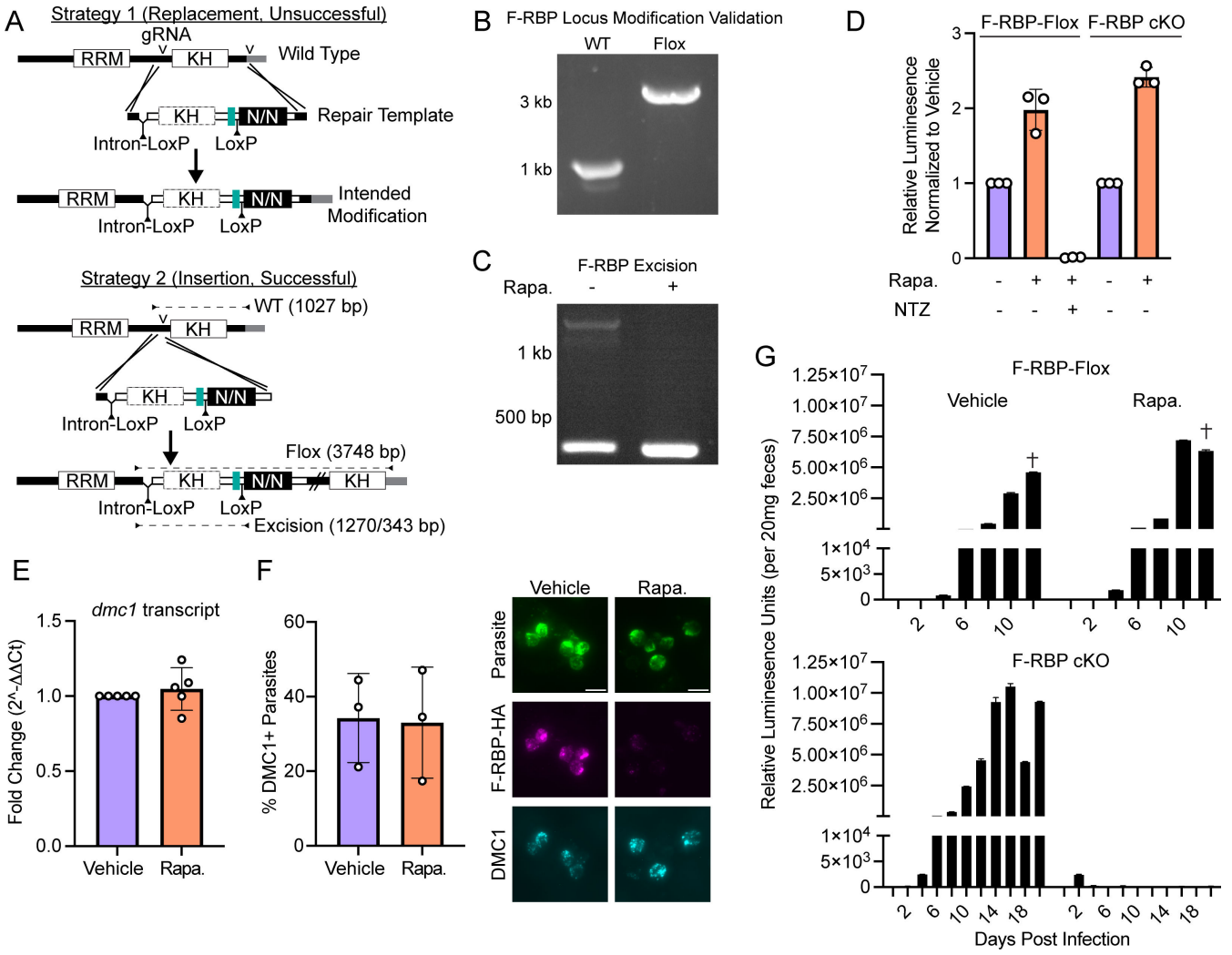


FIG 3 F-RBP is dispensable *in vitro* but reduces oocyst shedding *in vivo*. (A) Two strategies were attempted to insert loxP sites to generate an F-RBP cKO strain: replacement of the endogenous locus (top) and insertion of a new copy, creating a pseudo-diploid copy (bottom). Only the insertion strategy was successful. Further details are found in Fig. S1. Diagnostic PCR primer locations and lengths for panels B and C are shown in strategy 2 by dashed lines. gRNA locations are denoted above the gene model with arrows. The teal box is representative of a 3×HA tag, while N/N represents the nanoluciferase casette and neomycin selection marker. (B) Diagnostic PCR validating successful modification of the locus from wild type (left) to contain the floxed gene segment (right). (C) Cre-mediated excision of F-RBP in parasites grown for 48 h in HCT-8 cells with (right) or without rapamycin (left). Rapamycin is abbreviated as "rapa" in the remainder of figure legends. We note the excision in the absence of rapamycin, as previously reported (6, 7). (D) Parasite growth in HCT-8 cells after 48 h, measured by luminescence, for the parental F-RBP-flox strain (left) and F-RBP cKO strain (right). Both were treated with either vehicle or rapamycin, which are displayed in purple and orange, respectively, throughout. Raw luminescence values were normalized to the vehicle-treated condition within each genotype. Data are shown from three independent experiments, each performed in quintuplicate. Nitazoxanide (NTZ) treatment served as a negative control. (E) RT-qPCR measurement of *dmc1* transcript levels after 54 h of infection with F-RBP cKO parasites in HCT-8 cells. *C. parvum gapdh* was used as the control gene. Data are shown from five independent experiments, each conducted in triplicate. (F) Quantification of parasites staining positive for DMC1 after 54 h of F-RBP cKO infection in HCT-8 cells treated with vehicle or rapamycin. Note that DMC1 is present even when F-RBP is absent. Representative images are from one experiment, with 609 parasites counted across three coverslips per condition. Scale bar, 5 µm. (G) Oocyst shedding measured by fecal luminescence from *ifng*$^{-/-}$ mice infected with F-RBP-flox parasites (top) or F-RBP cKO parasites (bottom) treated with vehicle or rapamycin from Day −2 to 10 post infection. Data are representative of two independent experiments for F-RBP cKO infection and one experiment for F-RBP-flox infection. Three to five mice per experiment. Age and sex were matched within each experiment.

for 72 h, after which fecal matter was collected, the mice were sacrificed, and intestinal tissue was collected from the ileum of each animal for punch biopsies and histology (Fig. 4A). PCR using parasite genomic DNA extracted from the intestinal tissue as a template demonstrated that, at this time point, the *f-rbp* locus remained intact after

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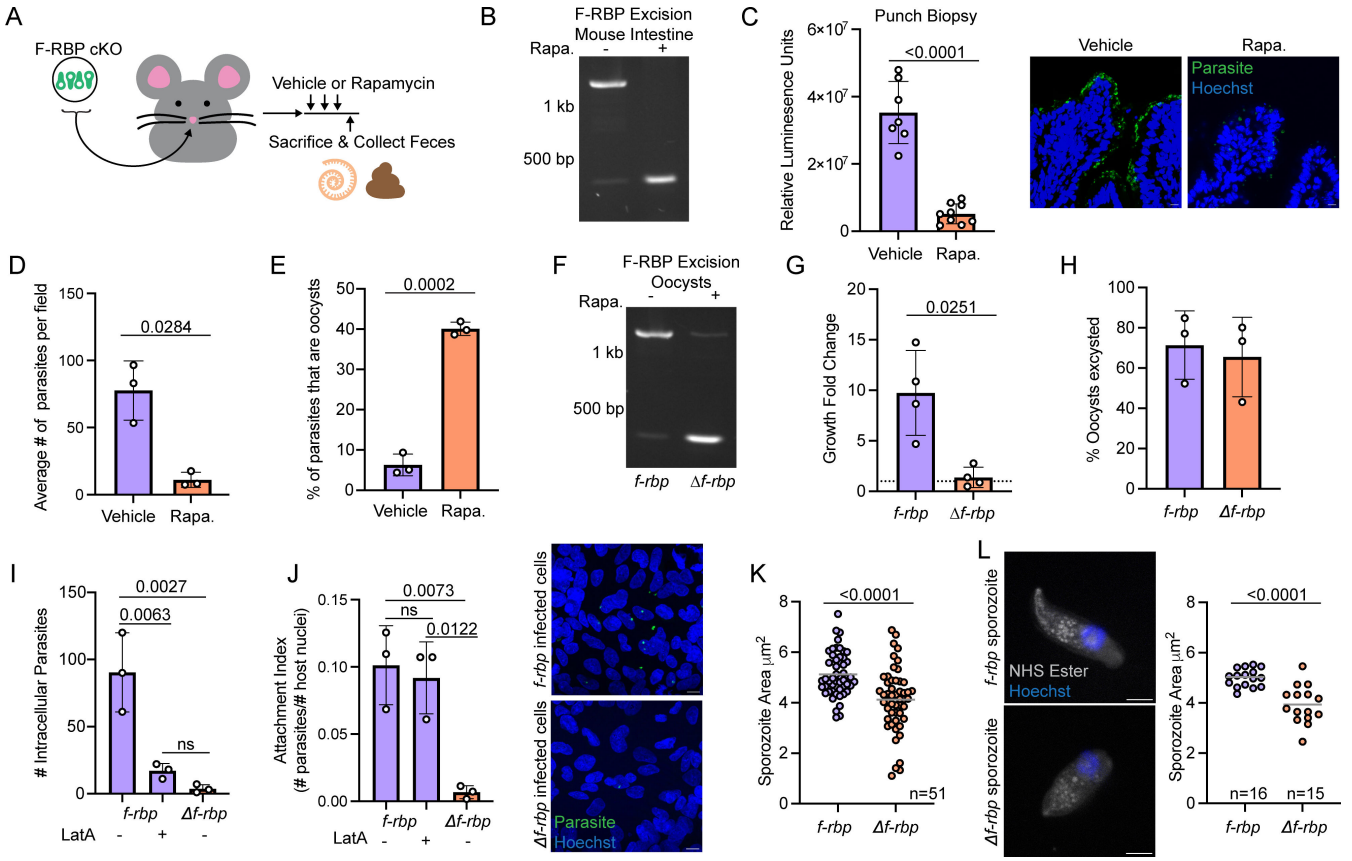


FIG 4 F-RBP-deficient female gametes give rise to sporozoites unable to infect host cells. (A) Experimental setup for assessing the absence of F-RBP *in vivo*. *Ifng*$^{-/-}$ mice were infected with F-RBP cKO parasites and allowed to grow for 6 days, after which the animals were treated with vehicle or rapamycin daily for 72 h. Feces were collected, and the ileums were harvested the next day. (B) Diagnostic PCR showing complete excision of F-RBP from parasites within the ileum in rapamycin-treated animals (right), compared with the vehicle-treated counterparts (left). Representative PCR results from two experiments; $n = 7$ and 9 for vehicle-treated and rapamycin-treated mice, respectively. (C) Luminescence measurements from ileal punch biopsies of each vehicle- or rapamycin-treated animal (left). Data are combined from two independent experiments. The same number of animals as in panel B was used. Three punch biopsies were collected per animal and measured in triplicate. The displayed value was calculated as the average for each biopsy. Welch's *t* test was used to determine significance. Representative images of parasite burden in an intestinal tissue section (right). Scale bar, 10 µm. (D) Number of parasites counted by immunofluorescence in intestinal tissue sections collected from animals after the treatment described in panel A. The average number of parasites counted per field of view for each animal (three animals per group) is displayed. A total of 1,230 parasites were counted across conditions. Welch's *t* test was used to determine significance. (E) Percentage of parasites counted in panel D exhibiting Crypto-Glo staining consistent with the oocyst stage. Welch's *t* test was used to determine significance. (F) Diagnostic PCR for F-RBP excision using DNA isolated from oocysts shed by mice infected with F-RBP cKO parasites and treated with vehicle or rapamycin. The resulting parasites shed by vehicle- and rapamycin-treated animals are referred to as *f-rbp* and Δ*f-rbp*, respectively. (G) Growth of *f-rbp* or Δ*f-rbp* parasites in HCT-8 cells after 48 h, measured by luminescence. Growth, measured as fold change, was calculated by dividing the average 48 h measurement by the average 4 h measurement for each genotype. Data are combined from four independent experiments, each measured in quintuplicate. Welch's *t* test was used to determine significance. (H) Percentage of *f-rbp* or Δ*f-rbp* oocysts with excysted sporozoites. Oocyst were counted and allowed to excyst, and the number of sporozoites was counted to determine the excystation percentage. Data are combined from three independent experiments. A total of 1,287 sporozoites were counted. (I) Quantification of intracellular parasites in HCT-8 cells 2 h after infection with *f-rbp* or Δ*f-rbp* parasites. An *f-rbp* sample was treated with latrunculin A (LatA) to inhibit invasion. Data are shown from one representative experiment out of three for *f-rbp* or Δ*f-rbp* and two experiments for *f-rbp* + LatA, with each experiment comprising of three coverslips. A total of 1,627 parasites counted. (J) Attachment index calculated from images collected in panel I as the number of parasites/the number of host cell nuclei (left). A total of 19,699 host cells counted. One-way ANOVA with multiple comparisons was used to determine significance for panels I and J. Representative images of fields of view containing HCT-8 cells infected with *f-rbp* or Δ*f-rbp* parasites (right). Scale bar, 10 µm. (K) Measured area in (µm^2) of *f-rbp* and Δ*f-rbp* sporozoites. Fifty-one parasites per group were measured. Welch's *t* test was used to determine significance. (L) Ultrastructure expansion microscopy (U-ExM) images of *f-rbp* or Δ*f-rbp* sporozoites (left). Note that Δ*f-rbp* sporozoites are smaller, stouter, and have a less well-defined apical end. Expansion factor, of 4.42. Scale bar, 1 µm. Sporozoite area in (µm^2) was also measured from U-ExM images. $n = 16$ for *f-rbp* and $n = 15$ Δ*f-rbp*. Welch's *t* test was used to determine significance.

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vehicle treatment but was fully excised following rapamycin treatment (Fig. 4B). We had assumed that the serious fitness defect and lower parasite burden in the intestine of rapamycin-treated animals (Fig. 4C) infected with the mutant were due to its inability to generate oocysts. We thus measured the number of oocysts and their proportion of the overall parasite population by immunofluorescence using suitable antibodies for both. As expected, we detected far fewer parasites in mice treated with rapamycin (Fig. 4D). Surprisingly, however, oocysts appeared overrepresented in the diminished parasite population after rapamycin treatment (Fig. 4E). We thus wondered whether ablation of *f-rbp*, while allowing oocyst production, renders those oocysts unable to initiate the next round of infection. To test this, we purified oocysts from the feces of groups of rapamycin-treated and control animals. We readily detected oocysts in the feces of both groups (albeit in lower numbers in the treated animals) and, importantly, documented robust ablation of the locus by PCR (Fig. 4F). *f-rbp* and Δ*f-rbp* oocysts were used to infect HCT-8 cell cultures, and nanoluciferase measurements were taken at 4 h and 48 h post infection to calculate parasite growth. *f-rbp* parasites showed growth, but *Δf-rbp* parasites did not (Fig. 4G). We conclude that loss of F-RBP leads to the production of oocysts unable to start the life cycle anew.

Δ*f-rbp* sporozoites are unable to invade host cells

We sought to determine the basis of oocyst sterility and first considered that *Δf-rbp* sporozoites may be unable to emerge from the oocyst. To initiate excystation, oocysts from both groups were resuspended in sodium taurodeoxycholate and sodium bicarbonate for 1 h (27), and excystation was scored. We found that both *f-rbp* and *Δf-rbp* oocysts were able to excyst with similar efficiency (Fig. 4H). We then assessed the ability of *Δf-rbp* sporozoites to invade host cells. Oocysts from both groups were excysted and allowed to interact with HCT-8 cells for 2 h, after which the cells were fixed and scored for successful invasion using a protection assay (28). We measured robust invasion by the vehicle-treated parasites, which was sensitive to the known inhibitor latrunculin (29) (Fig. 4I). In contrast, *Δf-rbp* sporozoites showed a profound invasion defect. Interestingly, we observed even fewer intracellular parasites with the mutant than with latrunculin treatment. Overall, we noted very few parasites following *Δf-rbp* exposure (Fig. 4I). Latrunculin prevents actin polymerization; treated parasites attach but then fail to glide and invade. We therefore measured parasite attachment and found *Δf-rbp* sporozoites unable to attach to host cells. Note that *f-rbp* sporozoites readily attach to their target, and this attachment is not affected by latrunculin treatment, as previously shown (29) (Fig. 4J).

We wondered whether these functional defects of the sporozoite were reflected in their morphology. We mounted *f-rbp* and *Δf-rbp* sporozoites onto poly-D-lysine slides, followed by staining with Hoechst and VVL. When observed by fluorescence microscopy and DIC, we noted that *Δf-rbp* parasites were significantly smaller than *f-rbp* sporozoites (mean areas, 4.123 μm^2 and 5.124 μm^2, respectively; $P < 0.0001$, Fig. 4K). By expansion microscopy, *Δf-rbp* sporozoites were again smaller (3.935 μm^2) and stouter, with their apical structure less defined than those of their *f-rbp* counterparts (5.031 μm^2) (Fig. 4L). We conclude that loss of F-RBP results in the shedding of oocysts that carry sterile sporozoites that are unable to attach to and invade epithelial cells to produce offspring by merogony.

F-RBP binds a subset of transcripts highly expressed in females

F-RBP is predicted to act as an RNA-binding protein based on the presence of a KH domain and an RRM(30). To test whether F-RBP binds RNAs and to identify its targets, we performed RNA Immunoprecipitation sequencing (31) using F-RBP-HA parasites. Western blot analysis of lysate from this strain subjected to anti-HA affinity matrix precipitation revealed a band of the appropriate size (90 kDa). This band is absent when wild-type lysate (WT) is used as control (Fig. 5A, IP fraction). HCT-8 cell cultures were infected

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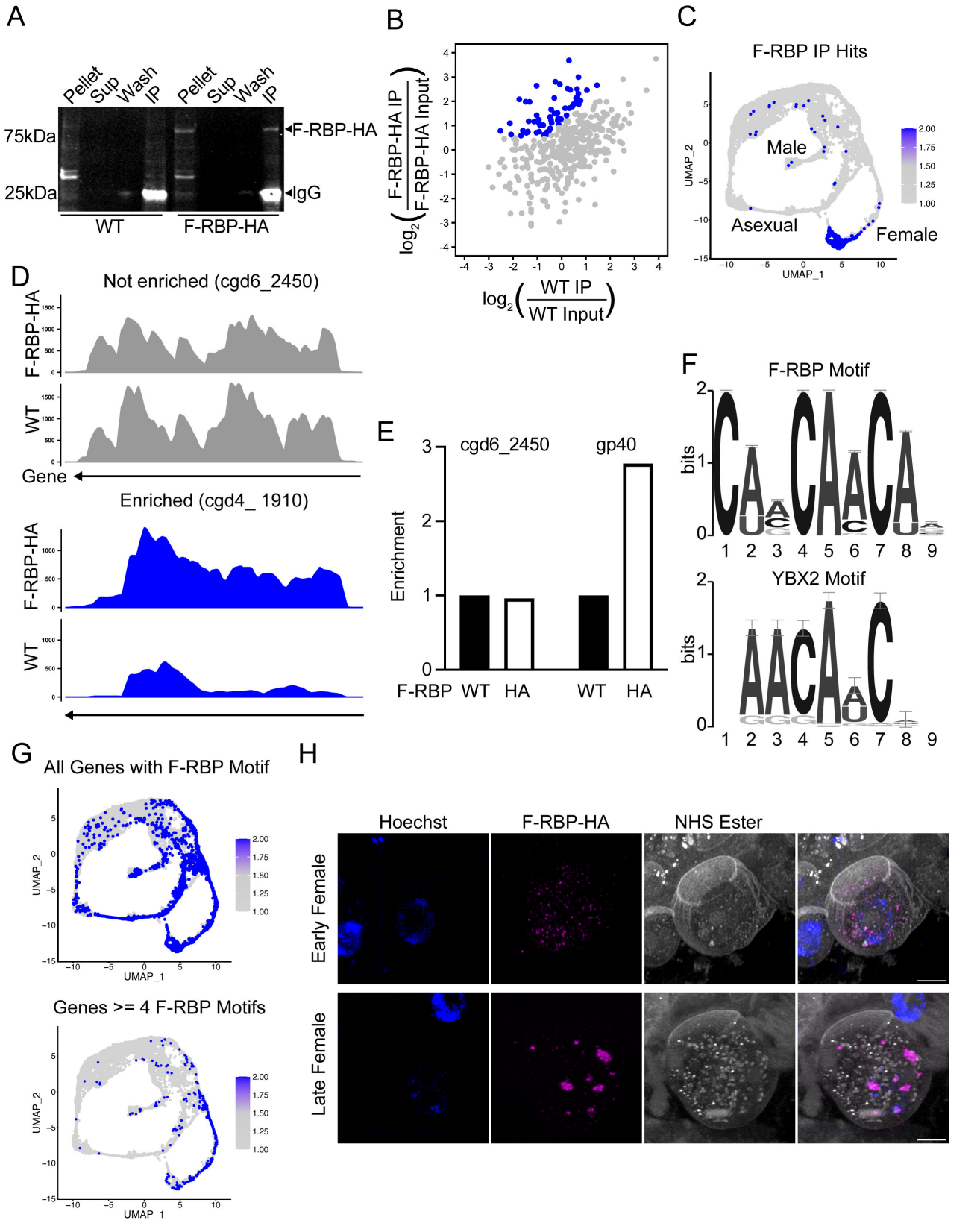


FIG 5 F-RBP binds specific transcripts upregulated in the female gamete and forms puncta in later female gametes. (A) Western blot detecting immunoprecipitation of F-RBP-HA from infected HCT-8 cells and a wild-type control using an anti-HA antibody after 54 h of infection. Lane annotations: Pellet (lanes 1 and 5) = cell pellet after sonication; Sup (lanes 2 and 6) = supernatant after IP, Wash (lanes 3 and 7) = washed affinity matrix before elution; IP (lanes 4 and 8) =

(Continued on next page)

Fig 5 (Continued)

immunoprecipitated sample. Note that the bright band at 25 kDa in lanes 1 and 8 represents the IgG light chain of the anti-HA antibody on the affinity matrix. (B) Relative enrichment of transcripts in the F-RBP-HA and WT samples. Relative enrichment is defined as the $\log_2$ ratio of IP to input transcripts per million (TPM). Blue dots represent transcripts enriched in F-RBP-IP compared with WT-IP. Non-enriched transcripts are represented in gray. (C) The 57 protein-coding RIP-seq-enriched transcripts are plotted on the Single Cell Atlas as a module. Details on module score computation can be found in the Materials and Methods. (D) Representative coverage plots for cgd6_2450 (a non-enriched transcript, top) and cgd4_1910 (an enriched transcript, bottom). Coverage is shown in counts per million. IP sample coverage plots are shown for both transcripts. The gene model and direction for each gene are represented by an arrow below the plot. (E) RT-qPCR measurements of a non-enriched transcript (cgd6_2450) and an enriched transcript (cgd6_1080, gp40). Measurements were taken from a biological replicate. Enrichment was calculated using the percent-input method. WT samples were normalized to an enrichment value of 1. (F) MEME Suite logos representing the F-RBP motif (CAMCAACAR, top) and the most similar motif, as determined by TOMTOM, RNCMPT00084, corresponding to that of the RNA-binding protein YBX2 (bottom). (G) All genes harboring the F-RBP motif identified by an unbiased scan of the *C. parvum* genome (top) and genes across the lifecycle with ≥4 copies of the F-RBP motif (bottom) are plotted as modules on the Single Cell Atlas. (H) U-ExM of female gametes in intestinal tissue sections. The earlier female (top) is identified by its smaller size and fewer wall-forming bodies than the later female (bottom). Note the poor labeling of the female nuclei due to high transcriptional activity, as previously reported (11). Expansion factor, 4.16. Scale bar, 1 µm.

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with F-RBP-HA parasites or control WT parasites, harvested after 54 h, and processed for immunoprecipitation and RNA sequencing. Reads were aligned to the *C. parvum* T2T genome (32), and, using a statistical cutoff based on the log odds, we determined that 57 parasite mRNAs and two lncRNAs were significantly enriched in the F-RBP-immunoprecipitated samples (Fig. 5B; Fig. S5A; Table S2).

Plotting the 57 protein-coding transcripts onto the *Cryptosporidium* Single Cell Atlas (6) showed enrichment for genes highly expressed late in the female gamete (Fig. 5C; Table S2), with 35 encoding proteins present in the sporozoite (Table S2) (12). Not all genes highly expressed in females were enriched in the IP. For example, cgd6_2450 shows high transcript abundance in females but is not enriched in the F-RBP IP (Fig. 5D) or in the RT-qPCR analysis of a biological replicate (Fig. 5E). However, the transcript of cgd4_1910 shows increased coverage in the F-RBP IP compared with the WT IP (Fig. 5D). Transcripts such as cgd6_1080, which encodes gp40 and shows enrichment by RIP-seq, also showed enrichment when RT-qPCR was used to probe pull-downs (Fig. 5E). We conclude that F-RBP binds a subset of mRNAs expressed in female gametes.

F-RBP targets share a sequence motif with targets of Y-box family proteins

We searched for shared primary-sequence features among the significantly enriched transcripts using the MEME Suite (33). This resulted in the detection of a CAMCAACAR motif in the coding sequence of 27 of the 57 of the enriched mRNAs (Fig. 5F; Table S3). A search for similar, previously identified eukaryotic motifs using the TOMTOM tool (34) returned 21 matches, which are shown in Fig. S6A and Table S3. Only one of these motifs showed significant enrichment, RNCMPT00084 corresponding to the motif recognized by Y-box binding protein 2 (YBX2), a member of the Y-box family (E-value and *q*-value <0.05, Fig. 5F; Fig. S6A; Table S3) (35). YBX2 is most notably associated with the development of gametes and early embryos (36, 37). Murine YBX2 is one of the most abundant maternal RNA-binding proteins (37) and a crucial mediator of oocyte RNA stability and translation (36, 38). An unbiased scan of the *C. parvum* genome for the CAMCAACAR motif found that 937 protein-coding transcripts contain at least one copy of this motif (Table S4). These genes appear more highly expressed at the end of merogony and female gamete development and sporogony (Fig. 5G). Among those that harbor multiple copies of the CAMCAACAR motif, female transcripts are enriched, and this enrichment is significant for the late female clusters 15-18 (Fig. 5G; Fig. S7A for more detail). Y-box proteins often aggregate into messenger ribonucleoprotein particles, RNA-protein complexes associated with mRNA stability and translational control (36). When we examined F-RBP localization at high resolution by expansion microscopy, we found a consistent developmental pattern. After 42h of HCT-8 infection in early female gametes, when the parasites were small and harbored few wall-forming bodies, F-RBP labeling was diffuse and distributed over many small particles across the cell (Fig. S8A, C,

and D). At 48h, when the gametes are larger with more wall-forming bodies, a significant decrease in the number of particles was observed; at the same time, these particles increased significantly in size (Fig. S8B through D). This pattern was also observed by expansion microscopy in tissue sections from infected mice where we see F-RBP aggregated into prominent cytoplasmic bodies in later females (Fig. 5H). We conclude that F-RBP interacts with mRNA, its targets are enriched for those carrying a conserved binding motif, and F-RBP forms aggregates akin to bodies described in other eukaryotes.

DISCUSSION

Infection with *Cryptosporidium* is a major cause of severe diarrheal disease around the world (39–41). The infection contributes to early childhood mortality and can result in malnutrition, stunting, and developmental delays (42–44). There is only one Food and Drug Administration-approved drug that lacks efficacy in those who need it most, immunosuppressed individuals and small children, and there are no vaccines available to protect children (45). The sexual phase of the parasite life cycle has emerged as a promising future target for intervention, as disruption of parasite sex rapidly halts infection (6, 11), which we corroborate in this study (Fig. 3G; Fig. S4A). Mechanistic understanding of parasite sex is required to identify specific molecular targets for such intervention (5). Abbreviating the infection in this way may stimulate immune-mediated protection, as prior exposure has been shown to lead to milder disease in subsequent infections (40).

Gamete development and fate determination are complex and highly dynamic processes in *Cryptosporidium,* a single-celled organism that lacks genetic sex determination in the form of mating types or sex chromosomes. Both male and female gametes are derived from the same asexual precursor cell with hard-wired timing (18). Evidence for circadian rhythms and the potential role of molecular clocks in protozoan parasites has been growing recently (reviewed in reference 46), and, along with cell cycle count and accumulation or depletion of a threshold factor, time is one of the conceivable drivers of the transition to gamete development. In the related apicomplexan parasite *Plasmodium*, the transcription factor AP2-G controls the initial commitment to sexual fate (20, 21), and the subsequent differentiation into male and female gametes involves both protein factors and non-coding RNAs (47, 48). In *Cryptosporidium*, a commitment factor akin to AP2-G is not apparent, but the transcription factor Myb-M is necessary and sufficient to initiate the male program (6). A reciprocal factor for the female gamete has not been discovered, and the female thus might be the default sex in the absence of Myb-M. Here, we identified the *f-rbp* gene among the earliest female transcripts. However, time-lapse imaging using a fluorescent reporter driven by the promoter of this gene demonstrated that it takes 5.5 h for the gene to be expressed, which represents almost the midpoint of the typical intracellular development cycle of this parasite. While fate commitment appears established when the merozoite leaves the last host cell (18, 49, 50), that fate is thus not immediately transcriptionally executed in the next host cell but takes hours to manifest (Fig. 6A; we acknowledge that future measurements with enhanced sensitivity might identify additional transcripts). In broader terms, this appears to be the case for all stages of *Cryptosporidium*, where the first half of each intracellular cycle is occupied by shared housekeeping tasks, including ribosome biogenesis and the establishment of an interface suitable for exporting parasite effector proteins and importing host metabolites (reviewed in reference 51). Only when these tasks are completed do three distinct transcriptional programs emerge from the shared origin: asexual merogony and male or female gametogenesis. The sex of the cell thus appears not as a primary overarching identity but as an alternative specialization halfway toward completion of intracellular development. We do not understand what defines fate prior to this point or what initiates the transcriptional transitions following it. In *Plasmodium*, a long non-coding RNA serves as a negative regulator of sexual commitment (52, 53). *Cryptosporidium* transcribes numerous long non-coding RNAs, many of which show stage-specific patterns of expression and may have yet-to-be-discovered regulatory

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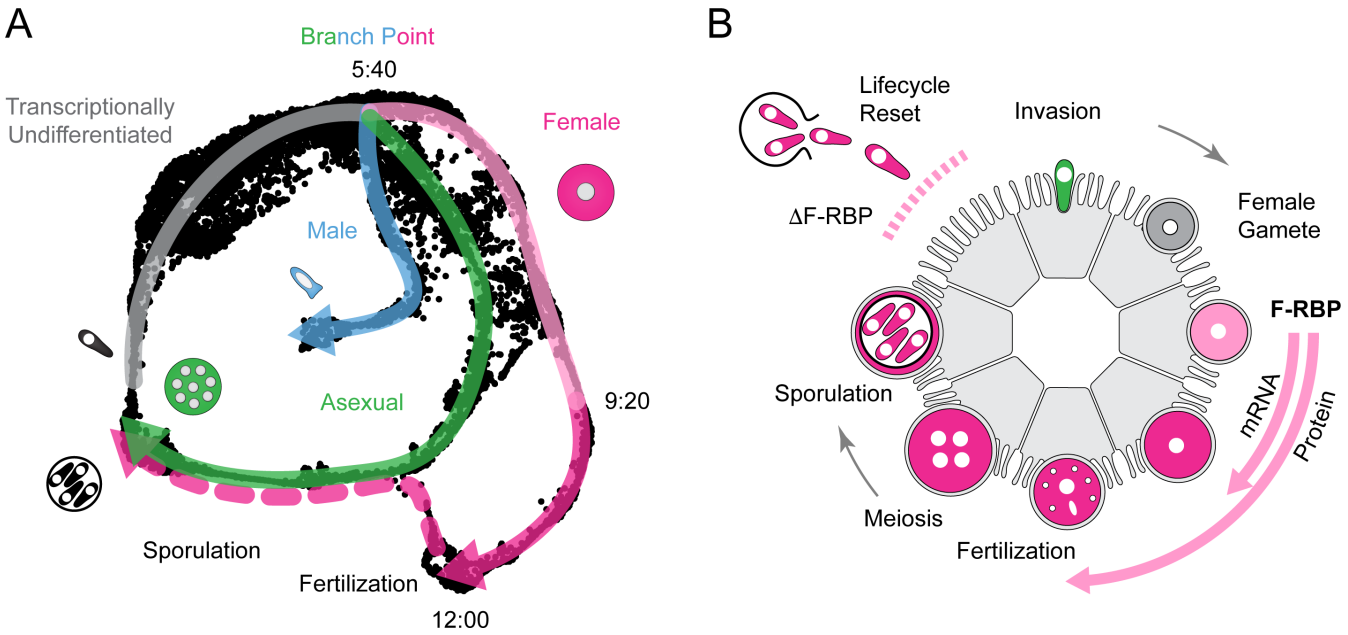


FIG 6 Transcriptional stage identity and F-RBP expression. (A) Life cycle model overlayed onto the Single Cell Atlas (6). "Real" time is indicated for female development based on the current study. Note that there is an extended initial period during which the stage does not appear to be transcriptionally manifest. Following the branch point, the expression of numerous stage-specific genes leads to differentiated merozoites and gametes. Late assembly of sporozoites (dashed pink line) resembles that of merozoites. (B) Schematic model of female development highlighting the presence of F-RBP mRNA and protein and the timing of the loss-of-function phenotype.

functions (6, 32, 54). Epigenetic regulation of promoter accessibility through dynamic histone modification offers another model, one that is important to the regulation of life cycle progression in other apicomplexans (55, 56).

While initiation of transcription is a pivotal point for regulating gene expression in apicomplexans, post-transcriptional events are also of critical importance. Multiple RNA-binding proteins were found to be critical to this process in *Plasmodium* and *Toxoplasma* (reviewed in references 57–59). Both rely on the Alba (acetylation lowers binding affinity) family of DNA/RNA-binding proteins to regulate developmental processes (60). Genome mining failed to identify Alba homologs in *Cryptosporidium*. Both *Plasmodium* and *Toxoplasma* express multiple proteins that contain RRM and KH domains, the two domains also found in F-RBP (30, 61).

RNA-binding proteins regulate the long-term storage and accurately timed sequential translation of mRNA, which are crucial to the development of gametes and embryos across the eukaryotic tree (1, 62). Numerous such proteins are active across gamete development and embryogenesis, and several have been studied in *Plasmodium* (reviewed in reference 58), often playing key roles in gamete development and fertility (47, 63). F-RBP is expressed very early in female development, but its loss does not cause defects in the development of female gametes (Fig. 6B). Fate choice, fertilization, and initial sporogony appear intact in Δ*f-rbp* mutants; however, the resulting oocysts contain sporozoites that are sterile (Fig. 4). This suggests that F-RBP influences development after fertilization. Several of the female RNA-binding proteins studied in *Plasmodium* similarly function downstream of sexual differentiation and in the development of the zygote. Among the most intensely studied in this group is the RNA helicase DOZI (development of zygote inhibited), which, in complex with its binding partner CITH (car-I and trailer hitch), represses the translation of transcripts in mature female gametocytes whose protein products are needed later for zygote formation and meiosis (3, 64). Also in this complex is 7-Helix-1, mutants lacking this factor show impaired gamete formation and reduced transmission of malaria parasites to mosquitoes (65). Puf2, another complex member, participates in translational repression. Several

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Puf2-controlled transcripts encode ookinete surface proteins (66), and in *P. yoelii*, Puf2 is important while the sporozoite awaits transmission in the salivary gland and is required for sporozoite infectivity (67). Ortholog and sequence similarity searches in *Cryptosporidium* reveal no homologs of Puf2 and 7-Helix-1. However, the *Cryptosporidium* genome does encode orthologs of DOZI and CITH (cgd8_1820 and cgd2_3220, respectively), and their transcripts appear present throughout most of the *Cryptosporidium* life cycle.

Our study identified a *Cryptosporidium*-specific example of a maternal RNA-binding protein required for the successful development of mature sporozoites and transmission to the next host. More specifically, we showed that F-RBP forms cytosolic granules in later-stage females, as was also observed for some of the proteins discussed above (3, 64, 65, 67). While many transcripts enriched by F-RBP immunoprecipitation encode sporozoite proteins present in organelles such as the dense granules, micronemes, and the crystalloid body, a significant number remained unassigned by spatial proteomic analysis (12). Interestingly, sequence analysis of the mRNAs enriched by F-RBP immunoprecipitation revealed significant enrichment of a motif matching that recognized by YBX2. YBX2 is a member of an ancient class of nucleic acid-binding proteins found in both prokaryotes and eukaryotes (68). In mammals, YBX2 is best known for regulating RNA homeostasis in gametes and early embryos and for assembling into cytoplasmic messenger ribonucleoprotein granules (36, 38, 69).

By what mechanism does maternal F-RBP ensure the production of infective sporozoites? More work is required to understand F-RBP function and to rigorously test a potential link between the phenotype observed here and binding of specific mRNAs. Based on precedent in other systems, we hypothesize that F-RBP is likely to regulate RNA homeostasis or translation. F-RBP may bind specific transcripts to influence their stability or decay. There are numerous examples of RNA-binding proteins acting in this fashion in oocytes (2, 4). In *Plasmodium*, Puf2 has been implicated in mRNA decay (66), while DOZI is thought to act in both stabilization and decay of mRNAs (70). F-RBP may also repress the premature translation of female transcripts needed later for sporozoite development. This is a well-documented strategy of post-transcriptional control used by egg cells (1) and also found to be active in *Plasmodium* female gametocytes (reviewed in reference 59), where misregulation of female translation was observed in the absence of Puf2 and DOZI (3, 66). mRNA homeostasis and translational control are not mutually exclusive mechanisms (66, 70), and it is likely that additional proteins are present alongside F-RBP in the granules we observe to aid in these tasks (3, 64, 65).

Ablation of F-RBP resulted in the first female-sterile mutant in *Cryptosporidium*. We demonstrate that successful control of gene expression throughout the complex life cycle depends not only on timely transcription initiation facilitated by different transcription factors but also on post-transcriptional processes that rely on stage-specific RNA-binding proteins and the formation of mRNA-protein granules.

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MATERIALS AND METHODS

Animals

Ifng$^{-/-}$ mice (RRID:IMSR_JAX:002287) were purchased from The Jackson Laboratory and propagated as a breeding colony at the University of Pennsylvania. Mouse housing conditions and age were as previously described (12-h/12-h light/dark cycle, 65°C–73°C , 5–8 weeks) (6).

Plasmid construction

All primers used in this study can be found in Table S1. Plasmids for genetic manipulation via CRISPR-Cas9 were generated as previously described (24, 71). Oligonucleotides (Sigma) encoding guide sequences were annealed and ligated into the *C. parvum* Cas9-U6 plasmid. Plasmids encoding the homology repair templates were made via Gibson assembly for the *f-rbp-mNeon-cowp1-tdTomato* reporter strain, restriction-ligation

cloning using AvrII and XmaI (New England Biolabs) for F-RBP-3xHA-eGFP, and BamHI and HindIII (New England Biolabs) for F-RBP-flox repair. Linear repair templates were generated by PCR using primers to encode recombination flanks with sequence identity to the target locus and PrimeSTAR Max DNA Polymerase Ver.2 (R047A, Takara).

Generation of transgenic parasites

C. parvum IOWA II strain (IIa subtype) oocysts were purchased from Bunchgrass Farms, and KVI parasites (IId subtype) (27) were maintained in-house through serial passage through *Ifng*$^{-/-}$ mice. Transgenic parasites were generated as previously described (24, 71, 72). Transgenic parasites in the Bunchgrass Farms genetic background were produced as described previously, including a 1-h excystation in 0.2 mM sodium taurodeoxycholate at 37°C and nucleofection using an Amaxa 4D nucleofector (Lonza) on program EH100 (6, 71). Transgenic parasites in the KVI background were produced as described previously (27), with a 1-h incubation in 10 mM HCl followed by a 1-h incubation in 0.2 mM sodium taurodeoxycholate and 20 mM sodium bicarbonate to promote excystation. *Ifng*$^{-/-}$ mice were treated with antibiotics 4–7 days prior to infection and gavaged with 8% sodium bicarbonate immediately before oral gavage with transfected sporozoites. Selection for stable transgenics and oocyst shedding were performed as previously described, with paromomycin selection in the drinking water at 16 g/L (24). Oocysts were purified from feces by sucrose flotation and cesium chloride gradient and stored in PBS at 4°C (24, 71, 72). Oocyst genomic DNA was extracted with the Quick-DNA Fecal/Soil Microbe Miniprep Kit (Zymo), and integration of the repair template was validated by PCR.

Cell line maintenance and preparation for infection

Human colorectal adenocarcinoma HCT-8 cells (ATCC, CCL-224) were maintained as previously described in RPMI-1640 supplemented with 10% fetal bovine serum (FBS), 1× penicillin-streptomycin, 1× gentamicin, and 1× fungizone (6). Bunchgrass Farms derived oocysts were prepared as described previously (71), with a 5-min incubation at 4°C in 1:4 bleach:PBS, centrifugation at 16,000 × *g*, and three washes with PBS, and then resuspended in infection medium (contains all components of complete RPMI-1640 with 1% FBS) and added to HCT-8 cells. KVI-derived oocysts were incubated for 1 h in 10 mM HCl at 37°C and washed twice with PBS, then excysted onto host cells in RPMI-1640 infection medium supplemented with 0.2 mM sodium taurodeoxycholate and 20 mM sodium bicarbonate.

Time-lapse fluorescence microscopy

HCT-8 infections were set up in eight-well chamber slides (CellVis). Movies were acquired on a GE DeltaVision OMX (Penn Vet Imaging Core) at 5% CO_2 and 37°C. Image stacks were acquired every 15 min for 26 h. Invasion to 15 h after invasion was analyzed to capture the developmental timepoints of interest. Movies were processed using GE softWoRx and Fiji. Fluorescence intensity was measured using Fiji for Mac OS.

Immunofluorescence assay (IFA)

For all HCT-8 infections

HCT-8 cells were grown in 96-well plates, washed, fixed at the time points indicated in the figure legends, and processed as previously described, including fixation in 4% PFA for 10 min, permeabilization with 0.25% Triton X-100 for 10 min, and blocking in 3% BSA for 1 h (6). Primary antibodies included rat anti-HA (1:1,000, Roche, 3F10), rabbit anti-HA (1:1,000, Cell Signaling Technology), VVL-Biotin (1:10,000, Vector Labs), VVL-fluorescein (1:1,000, Vector Labs, FL1231), and mouse anti-DMC1 (1:10, a gift from C. Huston, [8]). For the protection assay, cells were treated with latrunculin A (Abcam) at 2 µM, 30 min prior to infection. Parasites were allowed to infect for 2 h at 37°C prior to fixation. The staining

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protocol was adapted from reference 28. Cells were stained with 1:2,000 VVL-Biotin (Vector Labs) in 3% BSA for 30 min, then washed 3 times for 3 min with dPBS. Samples were permeabilized with 0.01% digitonin for 15 min and washed with PBS-0.01% Tween. For all HCT-8 experiments, Alexa Fluor secondary antibodies were used at a dilution of 1:1,000. Additional stains for the protection assay include Sporo-Glo-A600FLR (1:20 in Sporo-Glo buffer, Waterborne) for 1 h, followed by DAPI (1:1,000) for 10 min. Coverslips were mounted with Fluorogel (Electron Microscopy Sciences).

For intestinal sections

Mice were sacrificed either at the peak of infection or after the designated treatment time. Intestinal tissue was "Swiss-rolled" and processed as previously described by the Penn Vet Comparative Pathology Core (73). Sections were stained with the following primary antibodies: rat anti-HA (1:500, Roche, 3F10), Crypto-Glo-Biotin (1:20, Waterborne), and rabbit anti-sporozoite polyclonal antibody (1:1,000, a gift from F. Tosini [74]). Alexa Fluor secondary antibodies were used at a dilution of 1:250, with Hoechst at a concentration of 1:2,000. Slides were mounted as described in "For all HCT-8 infections," above.

For sporozoites

Parasites were excysted as described in "Generation of transgenic parasites," above, and fixed in suspension with 4% PFA. Sporozoites were stained in suspension with VVL-fluorescein (1:2,000,Vector Labs) and Hoechst (1:500) in PBS. Sporozoites were washed twice with PBS and transferred to an eight-well chamber slide coated with poly-D-lysine (CellVis). Sporozoites were allowed to settle for 20 min and were then washed with PBS and imaged.

For UEx-M

HCT-8 or intestinal samples were processed, sectioned, stored at −80°C, and expanded for U-ExM as described previously (75, 76). The primary antibody was rat anti-HA (Roche, 3F10), used at 1:250 for intestinal sections and 1:500 for HCT-8 samples. The secondary antibodies and stains used were Alexa Fluor goat anti-rat 488 (1:500), NHS ester Atto-647 (1:200, Sigma-Aldrich), and Hoechst (1:1,000). For sporozoites, parasites were excysted, transferred to poly-D-lysine-coated coverslips, and allowed to settle for 20 min. Sporozoites were spun at 200 × *g* for 1 min, and the coverslips were then processed as described above. The dyes used were Hoechst (1:1,000) and NHS ester Atto-647 (1:500, Sigma-Aldrich).

F-RBP-eGFP oocysts were imaged without fixation or staining after being allowed to settle on an Ibidi µ-Slide VI 0.4 (Thermo Fisher).

All images were acquired using a Leica DM6000 widefield microscope (Penn Vet Imaging Core) with a 100× objective or a Leica Stellaris FALCON 31594 confocal microscope (Penn Vet Imaging Core) using the 63× objective and Leica Application Suite X software. Images were processed and analyzed using Fiji for Mac OS software. For particle counting of U-ExM images, the area of interest was masked on the parasite using the NHS ester marker. All images were filtered with Gaussian blur with the auto settings; the threshold was set using the Otsu option with the auto settings, and a watershed filter was applied.

Flow cytometry

Oocysts were purified from feces using a miniaturized sucrose gradient and stained as previously described (25). Samples were processed on an LSRFortessa or a FACS Symphony A3 Lite. Data were analyzed with FlowJo software (Tree Star).

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Generation of loss-of-function mutant

To generate the F-RBP flox plasmid, a recodonized gene fragment was synthesized (GenScript) that included a multiple cloning site, an upstream endogenously encoded region of homology to F-RBP, an intron containing a loxP site, the recodonized portion of F-RBP starting at base 1723 including the KH domain but excluding the stop codon, a 3×-HA tag, and the downstream loxP site. This fragment was digested with BamHI and HindIII (New England Biolabs) and ligated into the *C. parvum* nanoluciferase-neomycin-resistant plasmid (24). The homology repair template and guide RNA were produced as described (24, 71). Animals were co-infected with 5,000 oocysts each of parasites harboring successful integration of this construct and parasites carrying the inducible Cre recombinase (25). Recombinant progeny were selected with paromomycin and BRD7929 as previously described (6, 25). Oocysts were purified from feces collected from days 18–24 post-infection and propagated in *Ifng*$^{-/-}$ mice.

Rapamycin treatment

For *in vivo* experiments, mice were treated with 10 mg/kg rapamycin (Thermo Fisher Scientific) or an equal dilution of ethanol, delivered by oral gavage, as previously described (6). Mice were treated daily by oral gavage from 2 days before infection to 10 days after infection. For *in vitro* experiments, parasites were cultured in the presence of 100 nM rapamycin or equally diluted ethanol as a vehicle control at the start of infection (6).

RT-qPCR

For whole cell lysate samples

Infection was performed in a 24-well plate. HCT-8 cells were infected and treated with vehicle or rapamycin as described in "Rapamycin treatment," above. RNA was isolated using the Qiagen RNEasy Kit, and cDNA was synthesized as previously described from 1µg RNA (6). cDNA was diluted 1:4 for all measurements. Reactions were conducted in SsoAdvanced Universal SYBR Supermix (Bio-Rad). The $\Delta\Delta C_t$ method (77) was used to determine relative expression, with *C. parvum gapdh* as the control gene. All qPCR primers have been published previously (78).

For RIP-seq validation

RNA Immunoprecipitation was performed as described in "RNA immunoprecipitation and sequencing (RIP-Seq)," below. RNA concentration was measured using a QuBit instrument before cDNA synthesis. 48 ng of RNA was used to generate cDNA. cDNA was produced as described for whole lysate samples. The percent-input method was used to determine enrichment between samples. Primers for gp40 (cgd6_1810) and cgd6_2450 have been published previously (78). RT-qPCR was performed using a ViiA 7 real-time PCR system (Thermo Fisher Scientific) with previously described conditions (6).

Nanoluciferase assay

Culture growth assay: HCT-8 cells were infected in quintuplet. At 4 h and/or 48 h post-infection, cells were lysed with lysis buffer (24) and mixed with Nano-Glo substrate (1:50; Nano-Glo Luciferase Assay Kit, Promega) in an opaque white 96-well plate. Luminescence was measured using a Promega GloMax plate reader (24, 71). From feces: 20 mg of feces was dissolved in 1 mL of fecal lysis buffer and processed as previously described (24). Percent reduction in oocyst shedding was calculated by dividing the area under the curve for the rapamycin-treated animals by the area under the curve for the paired vehicle-treated animals. From intestinal tissue: The ileum was removed, and three 5 mm punch biopsies were taken from the most distal region of the ileum from each

animal. Each sample was measured in triplicate, and the final values were calculated as the average of all samples for an individual animal.

gDNA extraction and PCR of intestinal tissue

An intestinal section measuring approximately 1 cm in length was collected from each animal. The tissue was digested, and DNA was extracted using the QIAmp DNA Mini Kit (Qiagen). DNA was diluted 1:5 to perform PCR.

Excystation assay

A total of 50,000 oocysts were excysted as described in "Generation of transgenic parasites," above. The number of oocysts was determined immediately after excystation was triggered using a hemocytometer (Kova Plastics). The number of resulting sporozoites was determined after 1 h of incubation at 37°C. Percent excystation was calculated by dividing the number of sporozoites by 4× the original oocyst count, assuming four sporozoites per oocyst.

RNA immunoprecipitation and sequencing (RIP-Seq)

Infections were set up in duplicate in T25 flasks 24 h before infection. In each flask, 4 × 10^6 Bunchgrass WT or F-RBP-HA parasites were used. The medium was changed to infection medium, and the flasks were incubated for 54 h. The RNA Immunoprecipitation protocol was modified from previous methods (79). Cells were fixed with 0.1% PFA for 8 min at room temperature. The reaction was quenched by adding glycine to a final concentration of 125 mM for 5 min. Cells were removed from the flask and pelleted at 500× *g* for 5 min at 4°C. The cells were washed twice with PBS and pelleted at 500× *g* for 5 min. Cells were resuspended in RNP lysis buffer (20 mM Tris, pH 7.5, 140 mM NaCl, 1.8 mM $MgCl_2$, 0.1% IGEPAL, 0.1% SDS, 10% glycerol, 1 mM DTT, 1× EDTA-free protease inhibitor cocktail [Roche], and 2 mM vanadyl ribonucleoside complex [NEB]). Cells were sonicated in three cycles of 10 s on and 30 s off at 20% power using a Sartorius Labsonic M sonicator. Sonicated samples were pelleted at 16,000× g for 30 min at 4°C. Ten percent of the supernatant volume was removed and frozen at −80°C for the input samples. Vanadyl ribonucleoside complex was added to the remaining supernatant, which was then used for immunoprecipitation with an HA-affinity matrix (Roche). Samples were immunoprecipitated overnight and processed as described in the product manual. RNA was isolated by incubation in RNA extraction buffer (20 mM Tris-HCl, pH 7.5, 5 mM EDTA, 50 mM NaCl, 0.1% SDS, and 50 µg/mL^{-1} proteinase K) at 70°C for 40 min. Samples of the same genotype were combined, and RNA was purified by phenol-chloroform extraction. Purified RNA was treated with DNase and purified again by phenol-chloroform extraction. This RNA was used to prepare cDNA-based, Illumina-compatible sequencing libraries using the Takara SMART-Seq Total RNA Pico Input with UMIs (ZapR Mammalian). The ribosomal cDNA depletion step of the SMART-Seq kit was omitted to minimize sample loss. Sequencing libraries were then converted into Ultima-compatible libraries using proprietary primers "UG-tR1 Indexing Primer v2, P1" and "UG-tR2 Universal Primer v2," provided by Ultima Genomics. Sequencing was performed on the UG 100 platform (Ultima Genomics), generating 130 to 160 million single-end reads per sample, with most reads ranging from 200 to 400 bp in length.

Reads were trimmed using Cutadapt (80) (-u 22, -a NNNNNNNNNNNNNNAGATCGGAAGAG, -q 20, -m 50, -l 500) and aligned to the *C. parvum* Bunchgrass telomere-to-telomere genome (32) (T2T-CpBGF, alignment to mRNA and lncRNA loci only) using STAR (81). Aligned reads were further processed using custom scripts that (a) remove ambiguous reads likely originating from rRNA loci nested within coding-gene loci and (b) subset BAM files to retain only unique alignments. The resulting cleaned unique alignments were used as input for salmon quant (82) (alignment-based mode, -l SR) to perform transcript quantification (transcript per million, TPM). Salmon output was then processed using a custom R script to identify putative F-RBP target transcripts.

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Only genes with more than five reads mapped in all four libraries were included in the analysis. Using TPM values for all remaining genes, we computed the base-2 logarithm of TPM^{IP}/TPM^{input} for both wild-type control and F-RBP-HA samples (see Fig. 5B) as a proxy for enrichment of each transcript in either pulldown. To identify putative F-RBP targets, we computed the base-2 logarithm of F-RBP-HA[TPM^{IP}/TPM^{input}]/WT[TPM^{IP}/TPM^{input}] and used the resulting F-RBP IP enrichment scores as input for mixture modeling with the R package mixtools (83) (normalmixEM, k = 2, maxit = 100,000, arbvar = F), as in previous studies leveraging it for RIP-seq analysis (84, 85). Mixture modeling in this case assumes that the observed enrichment scores arise from a combination of two underlying distributions, interpreted here as putative F-RBP targets (high enrichment score) and all other *C. parvum* transcripts (lower enrichment scores). The resulting Gaussian curves are shown in Fig. S5. mixtools assigns each transcript posterior probabilities of belonging to either underlying distribution. These probabilities are then used to calculate an odds ratio for each gene, allowing for model-based identification of putative F-RBP targets (LOD > 0). We further subset the resulting 86 candidates to include only genes with $\log_2$ (F-RBP-HA[TPM^{IP}/TPM^{input}]) > 0.5, to produce a finalized list of 59 putative F-RBP targets/IP hits.

For RIP-seq coverage plots (Fig. 5D), cleaned unique alignments were converted into CPM (counts per million)-normalized bigWig files using bamCoverage (deepTools; --normalizeUsing CPM, --binSize 1, --smoothLength 50). These bigWig files were then used as input for custom R scripts using the packages Gviz, tracklater, GenomicRanges, and IRanges to generate coverage plots.

For a detailed outline of RIP-seq data processing and analysis, refer to the supplemental code provided on GitHub (https://github.com/amdaniels22/Cryptosporidium_F-RBP_RIPSeqandMotif).

Motif analysis

The RIP-seq gene list was extracted from the T2T-CpBGF (32) genome annotation using AGAT v1.7.0 (86), and protein-coding genes were separated from lncRNA genes. Coding sequences (CDSs), 5′ UTRs, 3′ UTRs, 150 bp upstream regions, 150 bp downstream regions, and protein sequences were extracted from the mRNA annotations using the AGAT suite. All sequence types were prepared for motif-discovery analysis.

De novo motif discovery was performed using the MEME Suite v5.5.9 (33), with any number of repetitions per sequence (-mod anr). Separate runs were performed for DNA, RNA, and protein sequence. Only the CDS RNA motif model produced statistically significant motifs under the default *P*-value thresholds. The discovered motifs were compared using Tomtom from the MEME Suite.

The most significant motif model was scanned against the CDS sequences of all protein-coding genes in the T2T-CpBGF genome using FIMO from the MEME Suite. The output was parsed to count motif occurrences per gene, and a master data matrix was generated using a custom Python script. This matrix integrated (i) raw motif counts per CDS, (ii) gene lengths, (iii) motif counts normalized per 1 kb, (iv) mean FIMO *P*-values, *q*-values, and relative motif positions, and (v) presence/absence in the original RIP gene list. The resulting genes were stratified by motif count into four groups: one motif, two motifs, three motifs, and ≥4 motifs per CDS. Gene lists for each stratum were extracted from the master matrix using an awk script. We focused our analysis on genes containing ≥4 copies of this motif.

C. parvum single-cell RNA-seq data (GSE232438) were used to map motif-enriched gene sets to cells. A master module score was computed across all motif-containing genes (937 genes). Cells with master scores >0 were defined as the enriched cell set. Subgroup module scores were then computed for each motif-count stratum and masked to the enriched cell set (NA for non-enriched or subgroup-negative cells). UMAP feature plots were generated using the Seurat v5 package (87).

To investigate the cellular populations enriched for genes containing four or more copies of the discovered F-RBP motif, we performed cluster-based analysis using the

published single-cell data set (6). The absolute magnitude and proportion of cells enriched for motifs with ≥4 copies per cluster were visualized using raw-count and 100% stacked bar charts, respectively; the proportion visualization is shown in Fig. S7. To statistically validate cluster-specific enrichment, a one-tailed Fisher's exact test was performed for each cluster, comparing the numbers of enriched and non-enriched cells within that cluster with those in all other clusters combined. *P*-values were adjusted for multiple-hypothesis testing using the Benjamini-Hochberg false discovery rate method. Significance was defined as an adjusted *P*-value < 0.05.

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AUTHOR CONTRIBUTIONS

Abigail M. Daniels, Conceptualization, Formal analysis, Investigation, Visualization, Writing – original draft | Sebastian Shaw, Formal analysis, Investigation, Methodology,

Writing – review and editing | Christopher Nötzel, Formal analysis, Methodology, Writing – review and editing | Rui Xiao, Formal analysis, Methodology, Visualization, Writing – review and editing | Keenan O'Dea, Formal analysis, Investigation | Boris Striepen, Conceptualization, Formal analysis, Funding acquisition, Project administration, Supervision, Visualization, Writing – original draft.

DATA AVAILABILITY

RIP-seq data are available as an NCBI BioProject under accession number PRJNA1499133. The code used to process the RIP-seq data and perform motif analysis is available on GitHub (https://github.com/amdaniels22/Cryptosporidium_F-RBP_RIPSeqandMotif).

ETHICS APPROVAL

Animal experiments were conducted with approval from the Institutional Animal Care and Use Committee of the University of Pennsylvania under protocol #8066292.

ADDITIONAL FILES

The following material is available online.

Supplemental Material

Supplemental Figures (mBio01991-26-s0001.pdf). Figures S1–S8.
Supplemental Tables (mBio01991-26-s0002.xlsx). Tables S1–S4.
Movie S1 (mBio01991-26-s0003.mp4). Time-lapse of female development, showing two representative cells, using differential interference contrast (gray), F-RBP promoter-driven mNeon (green), and COWP1 promoter-driven tdTomato (red); the time stamp marks time after invasion shown in (hours:minutes:seconds).

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